

LAB 1 Inverter Chain

Technology: 22nm PTM-HP Nominal VDD = 0.8 V

WN = 44nm, WP = K · WN

L = 22nm for both PMOS and NMOS

7-stage inverter chain

HSPICE simulation sweeping K from 0.5 to 4.0 in steps of 0.1 (36 points), probing the out2 -> out3 stage.

Two parts below differ only in the input pulse rise/fall time X.

Part A — X = 10p

Deck: deck_7_inv.sp. Source data: results_10p/results.mt0; tabulated as results_10p/k_sweep_results.csv.

K at which tpHL == tpLH is 1.3

K at which tr == tf is 1.9

A.1 Propagation delay

tpHL and tpLH measured at the 0.4 V (50% of VDD)

K	tpHL (ps)	tpLH (ps)
0.50	2.558	4.403
0.60	2.695	4.320
0.70	2.768	4.062
0.80	2.740	3.761
0.90	2.832	3.553
1.00	2.914	3.396
1.10	2.995	3.285
1.20	3.065	3.210
1.30	3.136	3.134
1.40	3.212	3.083
1.50	3.292	3.060
1.60	3.376	3.059
1.70	3.462	3.053
1.80	3.549	3.050
1.90	3.639	3.025
2.00	3.730	3.005
2.10	3.822	2.993
2.20	3.916	2.986
2.30	4.010	2.981
2.40	4.096	2.978
2.50	4.182	2.979
2.60	4.269	2.976
2.70	4.355	3.039
2.80	4.442	3.013
2.90	4.528	3.052
3.00	4.615	3.061
3.10	4.726	3.060
3.20	4.835	3.062
3.30	4.939	3.074
3.40	5.039	3.067
3.50	5.136	3.093
3.60	5.230	3.093
3.70	5.321	3.098
3.80	5.410	3.120

K	tpHL (ps)	tpLH (ps)
3.90	5.500	3.124
4.00	5.591	3.130

A.2 Rise and fall time: tr, tf

tr and tf measured at out3, 10%-90% (rise) and 90%-10% (fall) of VDD (0.08 V <-> 0.72 V).

K	tr (ps)	tf (ps)
0.50	9.259	4.522
0.60	9.037	4.568
0.70	8.377	4.477
0.80	7.723	4.454
0.90	7.211	4.464
1.00	6.829	4.524
1.10	6.539	4.652
1.20	6.298	4.765
1.30	6.077	4.889
1.40	5.983	5.017
1.50	5.902	5.148
1.60	5.842	5.281
1.70	5.784	5.417
1.80	5.670	5.553
1.90	5.650	5.689
2.00	5.622	5.827
2.10	5.612	6.003
2.20	5.628	6.202
2.30	5.640	6.394
2.40	5.653	6.590
2.50	5.659	6.768
2.60	5.660	6.932
2.70	5.842	7.090
2.80	5.840	7.288
2.90	5.906	7.505
3.00	5.943	7.705
3.10	5.932	7.893
3.20	5.909	8.076
3.30	5.983	8.259
3.40	5.865	8.436
3.50	6.103	8.637
3.60	5.973	8.844
3.70	6.004	9.036
3.80	6.132	9.218
3.90	6.109	9.392
4.00	6.143	9.557

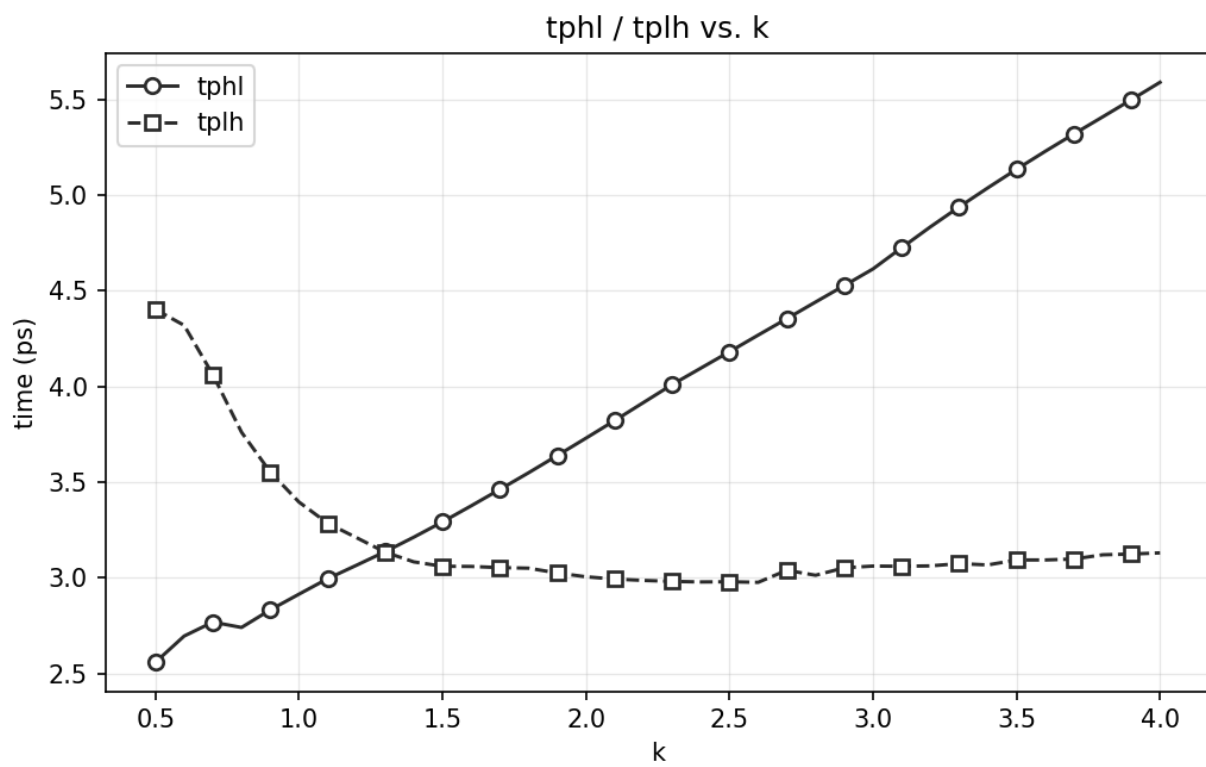


Figure 1: tpHL and tpLH vs K, X=10p

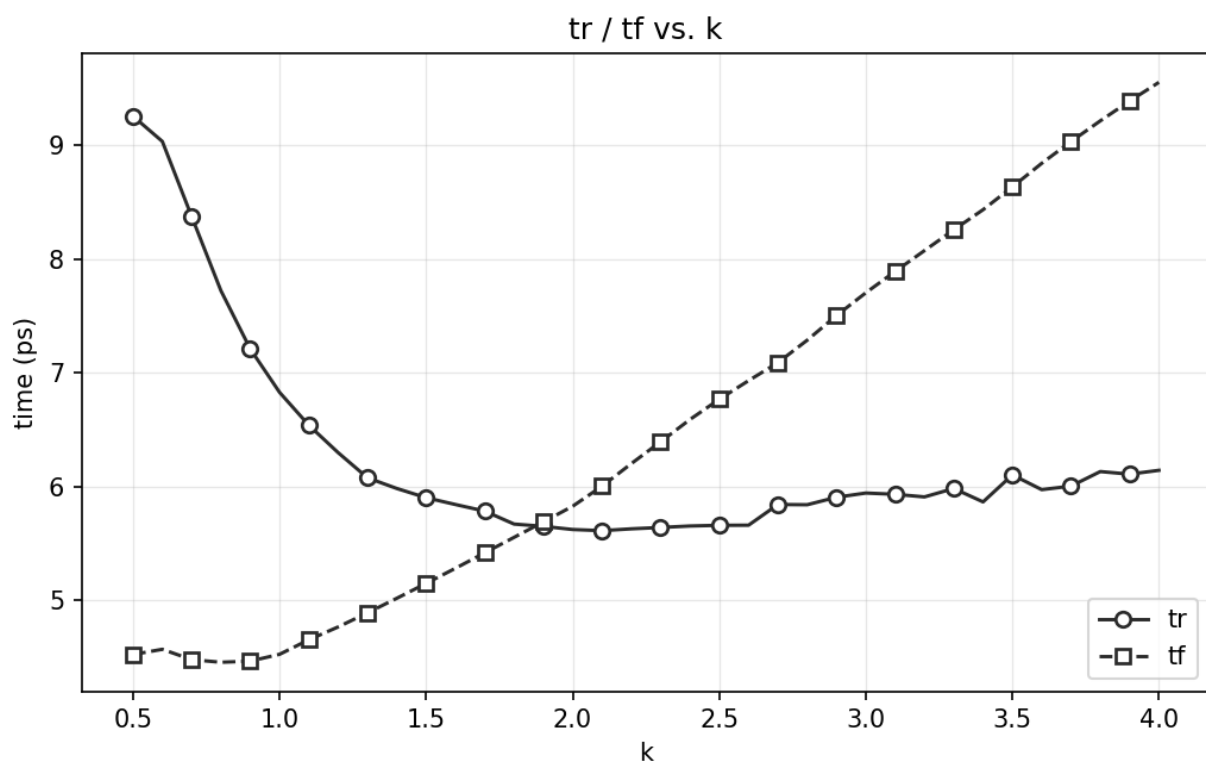


Figure 2: tr and tf vs K, X=10p

A.3 tpHL / tpLH vs. K

A.4 tr / tf vs. K

Part B — X = 5.65p

Deck: 1b_deck_7_inv.sp — same as Part A's deck_7_inv.sp, but with the input pulse rise/fall time X changed from 10p to **5.65p**. Source data: results_1b/results.mt0; tabulated as results_1b/k_sweep_results.csv.

K at which tpHL == tpLH is 1.33

K at which tr == tf is 1.9

B.1 Propagation delay: tpHL, tpLH

tpHL and tpLH measured at the 0.4 V (50% of VDD)

K	tpHL (ps)	tpLH (ps)
0.50	2.602	4.325
0.60	2.717	4.258
0.70	2.713	3.983
0.80	2.746	3.766
0.90	2.791	3.531
1.00	2.871	3.395
1.10	2.984	3.269
1.20	2.996	3.215
1.30	3.101	3.135
1.40	3.202	3.100
1.50	3.271	3.062
1.60	3.347	3.023
1.70	3.431	3.010
1.80	3.518	2.990
1.90	3.606	2.964
2.00	3.696	2.980
2.10	3.787	3.015
2.20	3.878	3.031
2.30	3.969	3.018
2.40	4.061	2.980
2.50	4.153	3.024
2.60	4.245	3.017
2.70	4.338	3.001
2.80	4.432	3.004
2.90	4.523	3.008
3.00	4.607	3.015
3.10	4.691	3.020
3.20	4.777	3.061
3.30	4.863	3.074
3.40	4.949	3.089
3.50	5.044	3.102
3.60	5.162	3.115
3.70	5.308	3.130
3.80	5.400	3.089
3.90	5.493	3.127
4.00	5.590	3.157

B.2 Rise / fall time: tr, tf

tr and tf measured at out3, 10%-90% (rise) and 90%-10% (fall) of VDD (0.08 V <-> 0.72 V).

K	tr (ps)	tf (ps)
0.50	9.166	4.406
0.60	8.997	4.455
0.70	8.426	4.517
0.80	7.726	4.437
0.90	7.174	4.440
1.00	6.780	4.487
1.10	6.497	4.644
1.20	6.213	4.573
1.30	6.086	4.719
1.40	5.992	5.013
1.50	5.922	5.151
1.60	5.839	5.295
1.70	5.793	5.442
1.80	5.761	5.589
1.90	5.741	5.738
2.00	5.743	5.887
2.10	5.749	6.034
2.20	5.724	6.187
2.30	5.673	6.395
2.40	5.761	6.599
2.50	5.749	6.797
2.60	5.694	6.991
2.70	5.635	7.181
2.80	5.672	7.368
2.90	5.706	7.552
3.00	5.736	7.711
3.10	5.760	7.889
3.20	5.837	8.074
3.30	5.949	8.261
3.40	6.032	8.449
3.50	6.094	8.625
3.60	6.138	8.798
3.70	6.214	9.029
3.80	6.037	9.195
3.90	6.115	9.352
4.00	6.335	9.505

B.3 tpHL / tpLH vs. K

B.4 tr / tf vs. K

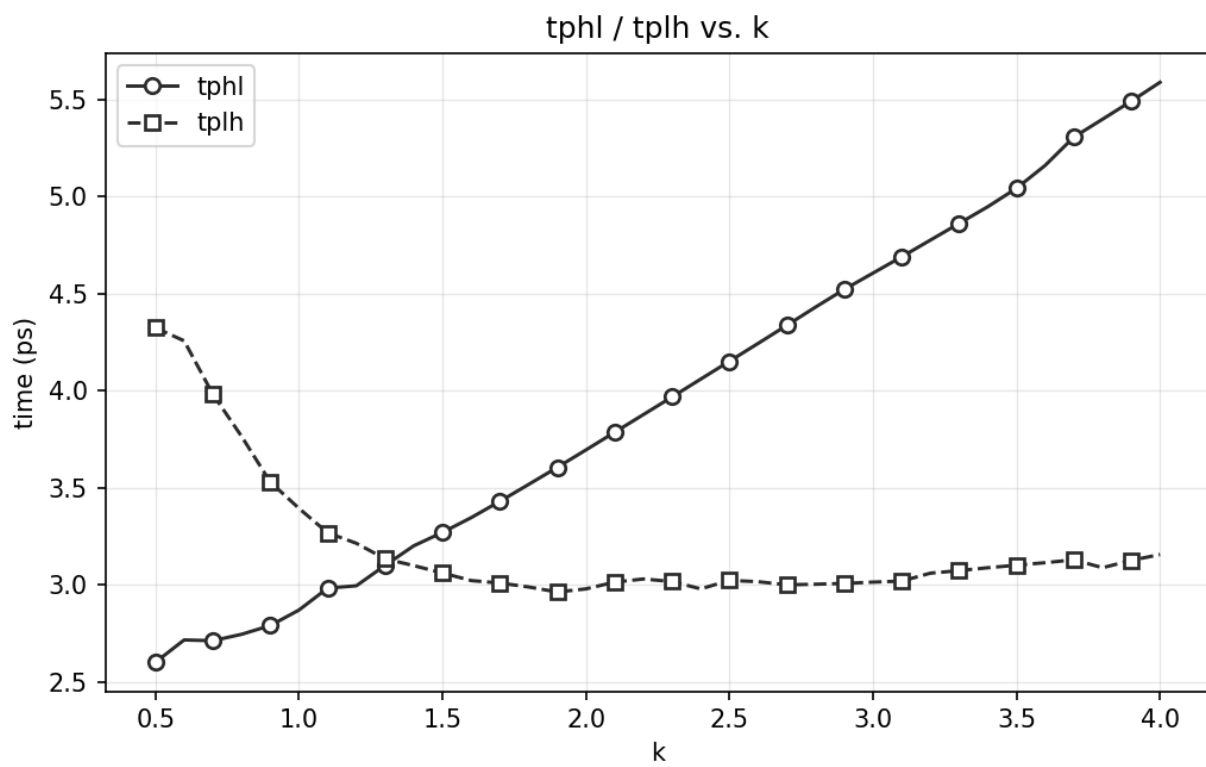


Figure 3: tpHL and tpLH vs K, X=5.65p

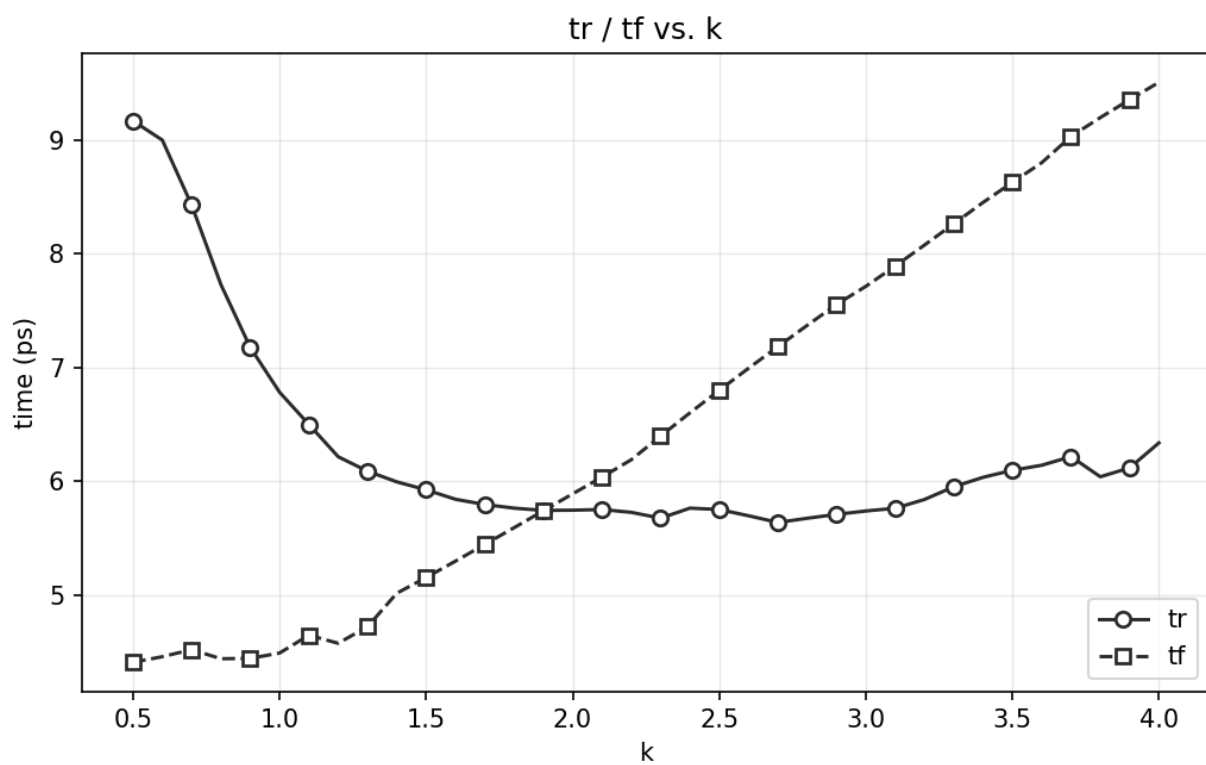


Figure 4: tr and tf vs K, X=5.65p